

VX/TPU — Change Log (V1.03-0)

Sector7 USA LLC · EDIT/TPU for Linux and Windows

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TPU Change Log

Product: Sector7 EDIT/TPU (`tpu.cpp`)

Current release: V1.03-0

Release date: 2026-07-17

Compared against: OpenVMS DECTPU **version=3 update=2** on host `t86` (OpenVMS x86_64 V9.2-3)

Release-numbering note: from this release TPU follows the same scheme as EDT — the middle field bumps once per release and the trailing edit field resets to 0. The earlier `V1.00-1` and `V1.00-2` builds are therefore renumbered here to `V1.01-0` and `V1.02-0`.

Bump `TPU_VERSION_*` / `TPU_RELEASE_DATE` / `TPU_SYSTEM_VERSION` / `TPU_SYSTEM_UPDATE` in `tpu.cpp` on every public release.

- **DECTPU-compatible:** `GET_INFO(SYSTEM, "version"|"update")` → integers
- **Sector7 extension:** `GET_INFO(0, "VERSION")` → `Vmajor.minor-edit` string

V1.03-0 — 2026-07-17

Licensing made optional + splash-once startup, from user field feedback.

Licensing / startup

- **S7SECURITY.INI is now OPTIONAL** — TPU runs whether or not a security file is present. The editor no longer exits with `tpu: security file not found` when the file is missing or unreadable.
- **Splash shown once, then suppressed** — on first launch (no `splash=off` flag present) the branded splash shows once (dismissed with RETURN), after which TPU writes a plain-text `S7SECURITY.INI` containing `splash=off` at the resolved path (`$S7SECURITY`, else `$HOME/S7SECURITY.INI`). Subsequent launches read that flag and skip the splash. Delete the file (or set `splash=on`) to see it again. `TPU_NO_SPLASH` still suppresses for a single run without persisting the flag.
- **Legacy license blob still honoured** — an existing encrypted `S7SEC1` blob is read tolerantly to preserve the licensed name/email on the splash, but it is no longer required and is replaced by the plain `splash=off` flag once the splash has been shown. `s7security/mksecurity` and `s7licdb` still generate encrypted blobs (now effectively single-use).
- Unlicensed splash drops the "licensed to ..." line and shows the plain `(c) Sector7 USA LLC TPU` banner.

V1.02-0 — 2026-07-16

Splash + documentation fixes from user field feedback.

Fixes

- **Splash no longer flashes by** — `show_splash_screen()` now waits for the user to press **RETURN** instead of auto-dismissing after a short timeout, so it no longer vanishes before it can be read. It first discards input buffered before it draws (e.g. the Enter used to launch `tpu`) with `flushinp()` so it isn't dismissed instantly; a single RETURN then clears it immediately so it never lingers annoyingly.

Docs

- **PDF code blocks** — reference/changelog `pre` blocks now use a light background instead of dark navy, fixing the unreadable "dark bricks" in Adobe Reader and on hard copy.

V1.01-0 — 2026-07-15

Linux regression hardening (RHEL8 / `rh8x`) + build-script link fix + required S7SECURITY license.

Features

- **S7SECURITY license blob** — Shared with EDT (`s7security/`). SHA-256 authenticated (static linked); locate via `$S7SECURITY` or `~/S7SECURITY.INI`. Splash line `(c) Sector7 USA LLC TPU licensed to <name> <email>` when `seconds > 0`; **seconds == 0 skips the splash**. Create files with `s7security/mksecurity`.

Fixes

- **DELETE(buffer) undo Cleanup use-after-free** — statement-exit undo Cleanup no longer touches a freed `Buffer*` after `DELETE`; clears `snap_buf`, wipes buffer/marker/range bindings, and verifies the buffer is still in `editor.buffers` before pop/clear of undo stacks. Glitch manifested on glibc as corrupted size vs. `prev_size / SIGABRT` during headless suites (macOS was more forgiving).
- **Linux link:** `run_tests.sh / run_regression.sh / run_coverage.sh` add `-ldl` on Linux so `dlopen/dlclose (CALL_USER / SPAWN paths)` link cleanly on RHEL8 (plain `-lcurses` alone failed with `DSO` missing from command line).

Tests

- Core regression green on macOS and `rh8x`: **104 / 35 / 134** passed (`tpu_tests / tpu_complete / tpu_coverage`) + callable smoke.
- Stress: 10× each headless suite on `rh8x` with no heap aborts.
- `run_tests.sh` exports `S7SECURITY` to `s7security/testdata/S7SECURITY.INI`.

V1.00-0 — 2026-07-15

First numbered public release identity + OpenVMS fidelity pass.

Language / builtins

- `GET_INFO(SYSTEM, "version"|"update")` DECTPU form (integers)
- `GET_INFO(0, "VERSION"|"PLATFORM"|"RELEASE_DATE")` Sector7 extension
- `MODIFY_RANGE(range, start, end)` with `LINE_BEGIN` / `LINE_END` / `BUFFER_*` keywords
- `ERROR` / `ERROR_LINE` / `ERROR_TEXT` last-error tracking
- `MARK(FREE_CURSOR)` + `SET (FREE_CURSOR)` virtual columns past EOL
- `TRANSLATE` range form
- `WRITE_FILE(buffer, filespec)` and range form (DEC-compatible)
- `FILE_PARSE` / `FILE_SEARCH` / `EXPAND_NAME` real Unix paths
- `PROMPT([prompt] [, default] [, NOECHO|ECHO])`
- `EXPAND_LEARN(learn)` → program value
- `SCROLL` / `SHIFT` viewport forms; `SPAWN` Linux shell; `CURRENT_WINDOW` `WINDOW` value
- `CONSTANT` reassignment → `TPU$CONTRADEF`
- `ATTACH` keyboard forward (Ctrl-Z detach / Ctrl-D EOF)
- Kill ring + `PASTE PREVIOUS`; named pastes `STORE_PASTE` / `RECALL_PASTE`
- Appendix B `MESSAGE_TEXT` catalog (~325 symbols in `tpu_messages_appendix_b.inc`)
- `CLI /COMMAND=` / `/SECTION=` (+ `--` forms); `SHOW VERSION` / `SHOW PASTES`
- Pattern expression compile cache (≤256)
- App.C probes: `GET_INFO(..., "BOUND"|"BEFORE_BOL"|"BEYOND_EOL"|"MIDDLE_OF_TAB")`

Tests / OpenVMS

- Regression baseline at release: headless + callable (`./run_tests.sh`)
- Cross-check host `t86`: DECTPU `GET_INFO(SYSTEM, "version")=3, "update"=2`
- Probe scripts: `t86/run_probe.sh`, `t86/run_crosscheck.sh` (shell Task / `required_permissions: ["all"]`)

Intentionally deferred

- RMS / `;n` file versions
- DECwindows Motif widget tree (accepted no-ops)
- Byte-identical VMS `.TPU$SECTION` / keystroke journal layouts (host uses portable `#VXSECT` + TPJB instead)
- `CONVERT` / global select / Motif

Arrays (DEC GET_INFO walk)

- `CREATE_ARRAY(n [, start])` static portion; typed integer indexes

- `GET_INFO(arr,`
`"FIRST"|"NEXT"|"LAST"|"PREVIOUS"|"CURRENT"|"LOW_INDEX"|"HIGH_INDEX"|"SIZE")`
- `GET_INFO(ARRAY, ...)` global array list
- `arr{k} := UNSPECIFIED / TPU$K_UNSPECIFIED; DELETE(arr)` wipes refs
- Identifiers may contain `$` (facility constants)

Also in V1.00-0 tree (post-identity)

- Section `SAVE / RESTORE / EXPAND + /SECTION= (#VXSECT)`
- Binary `TPJB` journals (text still recoverable)
- Callable API: `tpu_api.h`, `libtpu`, `TPU_Initialize` / `TPU$INITIALIZE ...`

Pre-V1.00 — baseline

Unversioned Sector7 TPU: language + EVE layer, patterns, FAO, journals as text scripts, `--test` harness, ncurses/PDCurses.

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